

Friendly Note (neighborly)

Italian people have a very peculiar way to permanently settle a dispute: they prepare a *friendly note*, politely describing their personal point of view about the matter, and deliver it secretly to each other. In order to add a dash of fun, they usually prepare the note by cutting and pasting sequences of characters from newspapers.

Giorgio makes no exception, and his long-standing dispute with his neighbour (about noise complaints) requires him to finally take action and defend his honour. He thus collected *a lot* of copies of his favourite newspaper, *La Stampa*, and is now ready to prepare his friendly note N .



Figure 1: A typical Italian friendly note.

Since Giorgio feels uncomfortable with scissors, he would prefer to cut and paste the smallest number of character sequences from the newspaper content S . Help Giorgio produce text N by juxtaposing **as few** contiguous subsequences of text S as possible, assuming that these subsequences can freely overlap (remember: he collected *a lot* of copies of that newspaper).

Among the attachments of this task you may find a template file `neighborly.*` with a sample incomplete implementation.

Input

The first line contains string N . The second line contains string S .

Output

You need to write a single line with an integer: the least number of subsequences Giorgio needs to cut and paste.

Constraints

- Strings N and S consist of ASCII printable characters only.
- $1 \leq \text{len}(N) \leq 1\,000\,000$.
- $1 \leq \text{len}(S) \leq 10\,000$.
- It is always possible to obtain the string N from substrings of S .

Scoring

Your program will be tested against several test cases grouped in subtasks. The score in each subtask will be calculated as the **minimum** score obtained in any of its test cases, multiplied by the value of the subtask. The score in a test case will be **0** if you output a number that is lower than the optimal solution or higher than the length of N . Otherwise, it will be calculated as:

$$\frac{\Delta_{\max} - \Delta_{\text{out}}}{\Delta_{\max} + 10\Delta_{\text{out}}}$$

where Δ_{out} is the difference between your solution and the optimal solution, and $\Delta_{\max}$ is the difference between the length of N and the optimal solution.

- **Subtask 1** [5 points]: Examples.
- **Subtask 2** [10 points]: N consists of a single character repeated multiple times.
- **Subtask 3** [20 points]: $\text{len}(N), \text{len}(S) \leq 20$.
- **Subtask 4** [25 points]: $\text{len}(N) \leq 1000$.
- **Subtask 5** [25 points]: $\text{len}(S) \leq 1000$.
- **Subtask 6** [15 points]: No additional limitations.

Examples

input.txt	output.txt
@dd10 vicin* 0 v@d1ci*n	6
caro vicino vuoi smetterla di fare casino Il caro vita sale ancora: se sei mancino vuoi smettere di comprare oggetti appositi, per esempio. Fare cassa e' sempre piu' difficile; parola di fiscalista.	5

Explanation

In the **first sample case**, the friendly note can be composed by the following fragments: "@d", "d1", "0 v", "1ci", "n", "*".

In the **second sample case**, the friendly note can be composed by the following fragments: "caro vi", "cino vuoi smetter", "la di f", "are cas", "ino".